Techno-Economic Assessment Framework for Wind-Assisted Propulsion System (WAPS) Retrofits on Tankers

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1 Introduction

This report documents the two coupled models used to assess whether retrofitting a tanker with a Wind-Assisted Propulsion System (WAPS) is cost-effective relative to a conventional engine-only vessel. Section 2 summarizes the steady-state kinematic model used to compute the engine thrust T_e required to hold a prescribed surge speed u_{ss} under given wind conditions (V_w, β_w) and a given sail installation. Section 3 presents the propagation of T_e through engine load, fuel consumption, and cost accounting to obtain the N-year discounted total cost of ownership. Section 4 provides representative vessel, sail, and economic parameters, together with a wind-climate probability model, for use in case studies. Curves and figures generated from the accompanying spreadsheet are included in 6.

2 Steady-State Thrust Model

2.1 Problem setup and assumptions

The full vessel model has six states X, Y, ψ, u, v, r (position, heading, surge, sway, yaw rate). For the purpose of estimating the required engine thrust, the surge speed is not solved for but is *prescribed* at the vessel's selected service speed, $u(t) = u_{ss} = u_d$, and the turning motion is assumed quasi-steady with constant heading, $r(t) = 0$. Physically, this corresponds to the sail array being positioned such that the rudder can be trimmed to exactly cancel the sail-induced yaw moment, so no net turning occurs. Under these two conditions the six-state system reduces to a two-equation nonlinear system in the unknowns

$$z = (v_{ss},\, \delta_{r,ss}),$$

the steady sway velocity and the steady rudder angle, with u_{ss} and the engine yaw moment N_e (taken as zero, i.e. steering is achieved by the rudder alone) held fixed as parameters. This system is solved once for every wind condition (V_w, β_w).

2.2 Apparent wind

For true wind speed V_w and direction β_w, the apparent wind seen in the body-fixed frame is

$$V_{ax} = V_w \cos\beta_w - u_{ss}, \tag{1}$$
$$V_{ay} = V_w \sin\beta_w - v, \tag{2}$$
$$V_a = \sqrt{V_{ax}^2 + V_{ay}^2}, \tag{3}$$
$$\beta_a = \operatorname{atan2}(V_{ay}, V_{ax}). \tag{4}$$

2.3 Sail aerodynamic forces

Consider a sail array of n sails, sail i having area $A_{s,i}$ and position (x_i, y_i) in the body frame. The lift coefficient $C_{L,i}$ depends on the selected sail technology:

- **Soft sail:** with sheet angle $\delta_{s,i}$ and angle of attack $\alpha_i = \beta_a - \delta_{s,i}$,

$$C_{L,i} = C_{L,\max} \sin(2\alpha_i).$$

- **Rotor sail (Flettner rotor, Magnus effect):** with prescribed spin ratio λ_i (a design/operating parameter, typically $\lambda_i \in [2, 4]$),

$$C_{L,i} = 2\lambda_i.$$

- **Suction sail:** with angle of attack $\alpha_i = \beta_a - \delta_{s,i}$, suction flow parameter σ_i, and empirical suction-augmentation coefficient $k_{s,i}$,

$$C_{L,i} = C_{L,\max} \sin(2\alpha_i) + k_{s,i}\,\sigma_i.$$

For every sail, independently of technology, the drag coefficient follows a quadratic polar

$$C_{D,i} = C_{D0} + k\,C_{L,i}^2,$$

with C_{D0} and k shared aerodynamic parameters. With dynamic pressure $q = \frac{1}{2}\rho_{\text{air}} V_a^2$, the lift and drag magnitudes are

$$L_i = q\,A_{s,i}\,C_{L,i}, \qquad D_i = q\,A_{s,i}\,C_{D,i}.$$

The resulting surge/sway force components and yaw moment of sail i, in the body frame, are

$$F_{x,i} = L_i \sin\beta_a - D_i \cos\beta_a, \tag{5}$$
$$F_{y,i} = L_i \cos\beta_a + D_i \sin\beta_a, \tag{6}$$
$$N_i = x_i F_{y,i} - y_i F_{x,i}. \tag{7}$$

The total sail force and moment follow by superposition,

$$F_x = \sum_{i=1}^{n} F_{x,i}, \qquad F_y = \sum_{i=1}^{n} F_{y,i}, \qquad N_s = \sum_{i=1}^{n} N_i. \tag{8}$$

2.4 Rudder force and moment

With rudder area A_r, longitudinal position x_r, lever arm l_r, and control angle δ_r, and effective inflow speed $u_{\text{eff}} = \max(|u_{ss}|, u_{\min})$, the rudder angle of attack (with $r = 0$) is

$$\alpha_r = \delta_r - \frac{v}{u_{\text{eff}}}.$$

With a linear lift law $C_{L,r} = 2\pi\alpha_r$, the rudder force and yaw moment are

$$F_r = \tfrac{1}{2}\rho_{\text{water}} A_r u_{ss}^2 C_{L,r}, \qquad N_r = x_r F_r. \tag{9}$$

2.5 Steady-state balance and solution

With u_{ss} fixed and $r = 0$, the sway and yaw dynamic equations reduce to the residual system

$$0 = F_y(u_{ss}, v_{ss}) + F_r(u_{ss}, v_{ss}, \delta_{r,ss}) - D_v\, v_{ss}, \tag{10}$$

$$0 = N_s(u_{ss}, v_{ss}) + N_r(u_{ss}, v_{ss}, \delta_{r,ss}) + N_e, \tag{11}$$

where D_v is the sway damping coefficient. Equations (10)–(11) are solved numerically (Newton-type root finding) for $z = (v_{ss}, \delta_{r,ss})$, for each wind condition (V_w, β_w).

2.6 Required engine thrust

The surge equation is not enforced during the solve above (since u_{ss} is prescribed rather than solved for); instead, once $(v_{ss}, \delta_{r,ss})$ are known, it is used directly to back out the engine thrust needed to sustain u_{ss}:

$$T_e(V_w, \beta_w) = D_u\, u_{ss}^2 - F_x(u_{ss}, v_{ss}), \tag{12}$$

where D_u is the surge damping coefficient, estimated from the calm-water resistance at 15 kn as

$$D_u = \frac{R_{@15\text{kn}}}{u_{15\text{kn}}^2}.$$

Since a negative value of (12) has no meaningful counterpart in the propulsive power/fuel accounting of Section 3 (it would indicate the sails alone already exceed the thrust needed at u_{ss}), the used value is clipped:

$$T_e^{\text{used}}(V_w, \beta_w) = \max\big(0,\, T_e(V_w, \beta_w)\big). \tag{13}$$

Evaluating (13) over the full (V_w, β_w) grid produces the thrust map that feeds the cost model below.

3 WAPS Life-Cycle Cost Model

3.1 Propulsive power and engine load

For each wind condition, the propulsive power and required engine power are

$$P_{\text{prop}}(V_w, \beta_w) = u_{ss}\, T_e^{\text{used}}(V_w, \beta_w), \qquad P_e(V_w, \beta_w) = \frac{P_{\text{prop}}(V_w, \beta_w)}{\eta_p}, \tag{14}$$

where η_p is the propulsive efficiency. The resulting engine load fraction is

$$\text{perc}_{\text{load}}(V_w, \beta_w) = 100\, \frac{P_e(V_w, \beta_w)}{P_{e,\max}}. \tag{15}$$

3.2 Fuel consumption

The engine's brake specific fuel consumption, $\mathrm{BSFC}(\mathrm{perc}_{\mathrm{load}})$ [g/kWh], is an engine-specific curve (from manufacturer data or a literature fit, typically exhibiting a minimum near 70–85% load). The instantaneous fuel consumption rate is

$$m_f(V_w, \beta_w) = P_e(V_w, \beta_w) \cdot \mathrm{BSFC}\big(\mathrm{perc}_{\mathrm{load}}(V_w, \beta_w)\big). \tag{16}$$

3.3 Wind-climate averaging

Let $\{V_{w,j}\}_{j=1}^{13}$ and $\{\beta_{w,k}\}_k$ be the discretized wind-speed and wind-direction grids (Section 4.4), and $P(V_{w,j}, \beta_{w,k})$ the joint probability of encountering that condition on the vessel's route, with $\sum_{j,k} P(V_{w,j}, \beta_{w,k}) = 1$. The expected (mean) fuel consumption rate is

$$\bar{m}_f = \sum_j \sum_k P(V_{w,j}, \beta_{w,k})\, m_f(V_{w,j}, \beta_{w,k}). \tag{17}$$

3.4 Annual fuel consumption and cost

With H the annual operating hours,

$$\Sigma FC = \bar{m}_f \cdot H, \qquad \Sigma C_f = \Sigma FC \cdot C_f, \tag{18}$$

where C_f is the fuel price (USD/kg).

3.5 Capital expenditure (CAPEX)

Given the equipment purchase cost ASSET (USD), and an installation cost equal to 25% of ASSET for a previously non-hybrid vessel,

$$\mathrm{CAPEX} = \mathrm{ASSET} + \mathrm{INSTALL} = 1.25 \cdot \mathrm{ASSET}. \tag{19}$$

3.6 Operating expenditure (OPEX)

The annual operating expenditure is

$$\mathrm{OPEX}_{\mathrm{annual}} = 0.02 \cdot \mathrm{CAPEX} + C_{\mathrm{el}}, \tag{20}$$

where C_{el} is the annual electrical operating cost of the WAPS installation (e.g. rotor motors, suction fans).

3.7 Emissions compliance cost

The annual CO_2/CO emission compliance cost is estimated from the annual fuel burn,

$$\mathrm{Cost}_{\mathrm{emiss}} = \mathrm{EF}_{\mathrm{CO_2}} \cdot \Sigma FC \cdot P_{\mathrm{CO_2}} \cdot \varphi, \tag{21}$$

where $\mathrm{EF}_{\mathrm{CO_2}}$ is the fuel's CO_2 emission factor (tCO_2 per tonne fuel), $P_{\mathrm{CO_2}}$ is the effective carbon price (USD/tCO_2), and $\varphi \in [0, 1]$ is the fraction of the vessel's annual fuel consumption exposed to carbon pricing on its trading routes. Because ΣFC is lower for a WAPS-equipped vessel, (21) is correspondingly lower than for an engine-only vessel of the same duty.

3.8 Total annual cost and N-year life-cycle cost

The total annual cost is

$$C_{\text{annual}} = \Sigma C_f + \text{OPEX}_{\text{annual}} + \text{Cost}_{\text{emiss}}. \tag{22}$$

The N-year total cost of the WAPS installation, including CAPEX and the discounted stream of annual costs at discount rate r, is

$$C_{\text{WAPS}}(N) = \text{CAPEX} + \sum_{\text{year}=1}^{N} \frac{C_{\text{annual}}}{(1+r)^{\text{year}}}. \tag{23}$$

Comparing $C_{\text{WAPS}}(N)$ across sail technologies, configurations, or against an engine-only baseline (CAPEX = 0, higher ΣFC and $\text{Cost}_{\text{emiss}}$) as a function of N gives the year-by-year cost curves used to judge cost-effectiveness.

4 Representative Data for Case Studies

All values below are representative/illustrative and intended as reasonable defaults for a first case study; they should be refined with vessel- and route-specific data where available.

4.1 Reference vessel and hydro/aero model parameters

Table 1 lists the parameters entering the thrust model of Section 2, consistent with a large tanker (Aframax/Suezmax class).

Table 1: Reference vessel and force-model parameters.

Parameter	Symbol	Value
Mass	m	$4.5 \times 10^7\,\text{kg}$
Yaw moment of inertia	I_z	$3.5 \times 10^{11}\,\text{kg m}^2$
Length	–	220 m
Beam	–	40 m
Draft	–	14.5 m
Air density	ρ_{air}	$1.225\,\text{kg/m}^3$
Water density	ρ_{water}	$1025\,\text{kg/m}^3$
Rudder area	A_r	$55\,\text{m}^2$
Rudder longitudinal position	x_r	-60 m
Rudder lever arm	l_r	4 m
Sail-array longitudinal position	x_s	2 m
Surge damping	D_u	$9645\,\text{kg/m}$
Sway damping	D_v	$3.0 \times 10^6\,\text{kg/s}$
Yaw damping	D_r	0
Base aerodynamic lift/drag coeffs.	$C_{L,\max}$, C_{D0}, k	1.2, 0.05, 0.12

4.2 Sail technology parameters

Table 2 gives representative per-sail parameters for each technology, for an array of $n = 4$–6 units.

Table 2: Representative sail technology parameters.

Technology	Parameters	Representative range
Soft sail	Area A_s per sail	100–250 m^2
	Sheet angle δ_s	0–90° (operated/controlled)
Rotor sail (Flettner)	Radius R_s	1.5–2.5 m
	Height (implicit in area)	24–35 m
	Spin ratio λ	2–4
Suction sail	Area A_s per sail	100–200 m^2
	Suction coefficient k_s	8–15
	Suction flow parameter σ	0.10–0.20

4.3 Techno-economic parameters

Table 3 gives representative vessel operating and cost parameters for an Aframax-class tanker; Table 4 gives representative WAPS cost parameters (rotor-sail figures reflect current commercial retrofit experience).

Table 3: Representative vessel operating/economic parameters.

Parameter	Symbol	Representative value
Deadweight	DWT	115 000 t
Design/service speed	u_d	14 kn (≈7.2 m/s)
Resistance at 15 kn	$R_{@15\mathrm{kn}}$	0.9–1.0 MN
Maximum engine power	$P_{e,\max}$	12–14 MW
Propulsive efficiency	η_p	0.65–0.70
Annual operating hours	H	6500–7000 h
Fuel price (VLSFO, 2026)	C_f	0.60–0.80 USD/kg
CO_2 emission factor	$\mathrm{EF}_{\mathrm{CO_2}}$	3.114 tCO_2/t fuel
Effective carbon price	$P_{\mathrm{CO_2}}$	80–100 USD/tCO_2

Table 4: Representative WAPS capital/operating cost parameters.

Item	Symbol	Representative value
ASSET cost per rotor sail	–	0.4–1.0 million USD
ASSET cost per soft/suction sail	–	0.3–0.8 million USD
Installation cost (fraction of ASSET)	–	25%
Annual OPEX (fraction of CAPEX)	–	2%
Annual electrical operating cost	C_{el}	50 000–150 000 USD
Discount rate	r	6%–8%

For reference, a documented two-rotor retrofit on a similarly-sized product tanker reported fuel savings on the order of 7–10% on typical trading routes, illustrating the expected order of magnitude of ΣFC reduction relative to an engine-only baseline.

4.4 Wind climate: discretization and probability matrix

The wind-speed and wind-direction grids are

$$V_{w,j} \in \{0, 2, 4, \ldots, 24\} \text{ m/s} \quad (13 \text{ values}),$$

$$\beta_{w,k} \in \{0, 20, 40, \ldots, 180\}^\circ \quad (10 \text{ values, step } 20^\circ).$$

This direction grid contains 10 values.

Route- and season-specific joint probabilities $P(V_w, \beta_w)$ require operational data (e.g. hindcast/route climatology) and are not derived here. As a placeholder, an illustrative joint distribution is constructed under an independence assumption,

$$P(V_{w,j}, \beta_{w,k}) = f_V(V_{w,j}) \cdot f_\beta(\beta_{w,k}), \tag{24}$$

with a Rayleigh (Weibull, shape $k = 2$) marginal for wind speed with scale $c = 8\,\text{m/s}$,

$$f_V(V_w) = \exp\Big[-\big(\tfrac{V_w - 1}{c}\big)^2\Big] - \exp\Big[-\big(\tfrac{V_w + 1}{c}\big)^2\Big],$$

(bin-integrated over each $\pm 1\,\text{m/s}$ speed bin, renormalized to sum to 1), and a $\sin\beta_w$-weighted marginal for direction (favoring beam/quartering encounters over dead-ahead/dead-astern),

$$f_\beta(\beta_w) = \frac{\sin\beta_w}{\sum_k \sin\beta_{w,k}}.$$

Tables 5 and 6 give the resulting marginals on the grids above.

Table 5: Illustrative wind-speed marginal $f_V(V_w)$ (Rayleigh, $c = 8\,\text{m/s}$).

V_w [m/s]	0	2	4	6	8	10	12
f_V	0.0155	0.1157	0.1922	0.2116	0.1830	0.1310	0.0796

V_w [m/s]	14	16	18	20	22	24
f_V	0.0417	0.0188	0.0074	0.0025	0.0008	0.0003

Table 6: Illustrative wind-direction marginal $f_\beta(\beta_w)$ ($\sin\beta_w$-weighted).

β_w [°]	0	20	40	60	80	100	120	140	160	180
f_β	0.000	0.060	0.113	0.153	0.174	0.174	0.153	0.113	0.060	0.000

The joint probability matrix used in (17) is then the outer product of Tables 5 and 6 via (24); each row/column should be replaced with route-specific climatology when available.

5 Case Study

To verify the cost estimation methodology described, we have established a comparison study for 8 retrofit WAPS examples and a no-sail (engine only) baseline on a 45MT medium product tanker (Suezmax). The vessel construction, engine and functional parameters are presented in the Table of Fig. 1. The 8 WAPS configuration details for each one of the corresponding examples are described in the Table of Fig. 2. The examples studied include 4 cases of soft sail installation (with $120m^2$ or $180m^2$ area per sail with a total of 4 to 6 soft sails), 2 cases of rotor sails (2 or 4 respectively) and 2 cases of sunction sail installation ($150m^2$ for each one of 4 or 6 sunction sails).

Vessel database			
Terminology	**Symbol**	**Units**	**Cases 1-9**
Vessel ID	—	—	1
Vessel type	—	—	Medium Product Tanker
Deadweight	DWT	t	115000
Displacement/mass	m	kg	45000000
Length	L	m	220
Beam	B	m	40
Draft	T	m	14.5
Design/service speed	Ud	m/s	7.2
Resistance @ 15 kn	R15	MN	0.5
Maximum engine power	Pe,max	W	12000000
Propulsive efficiency	ηp	%	70
Annual operating hours	tyr	h	6570
Fuel price	Cf	USD/kg	0.8

Figure 1: Vessel Properties Baseline (same for all cases).

Sail Configuration			**Case 1**	**Case 2**	**Case 3**	**Case 4**	**Case 5**	**Case 6**	**Case 7**	**Case 8**	**Case 9**
Technology			None	Soft Sail	Soft Sail	Soft Sail	Soft Sail	Rotor Sail	Rotor Sail	Sunction Sail	Sunction Sail
Number of sails	Ns	-	-	4	4	6	6	2	4	4	6
Area per Soft/Sunction sail	As	m^2/sail	-	120	180	120	180	-	-	150	150
Total Soft/Sunction Sails area	Ns*As	m^2	-	480	720	720	1080	-	-	600	900
Sail x-positions	x_s	m	-	-80, -60 m	-80, -60 m	-70, -60, -50 m	-70, -60, -50 m	-70m	-80, -60 m	-80, -60 m	-70, -60, -50 m
Sail y-positions	y_s	m	-	-10, +10 m	-10, +10 m	-10, +10 m	-10, +10 m	-10, +10 m	-10, +10 m	-10, +10 m	-10, +10 m
Sail control			-	optimized	optimized	optimized	optimized	optimized	optimized	optimized	optimized
Rotor sail radius	Rs	m	-	-	-	-	-	2	2	-	-
Rotor sail height	H_rotor	m	-	-	-	-	-	28	28	-	-
Rotor sail spin ratio	λ	-	-	-	-	-	-	2	2	-	-
Sunction sail coefficient	ks		-	-	-	-	-	-	-	12	12
Sunction sail parameter	σ		-	-	-	-	-	-	-	0.15	0.15

Figure 2: Sail Configuration for each WAPS case study.

6 Results

The annual/N-year discounted cost curves $C_{\text{WAPS}}(N)$, compared across sail technologies and against the engine-only baseline is shown in Fig. 3 for each particular year and in Fig. 4 particularly for the first 10, 20 and 25 years. Furthermore, the Table in Fig. 5 presents the annual fuel, compliance and total cost for each particular example. (Discussion neeeded).

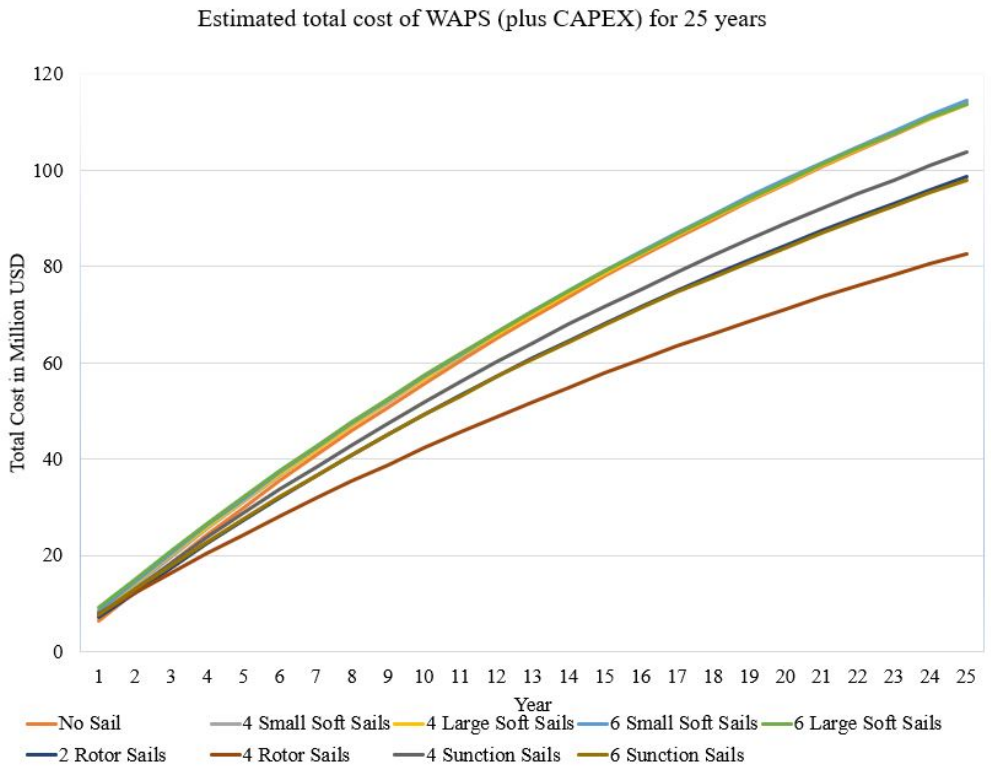


Figure 3: Long-term 25-year Cost Prediction for each WAPS case study.

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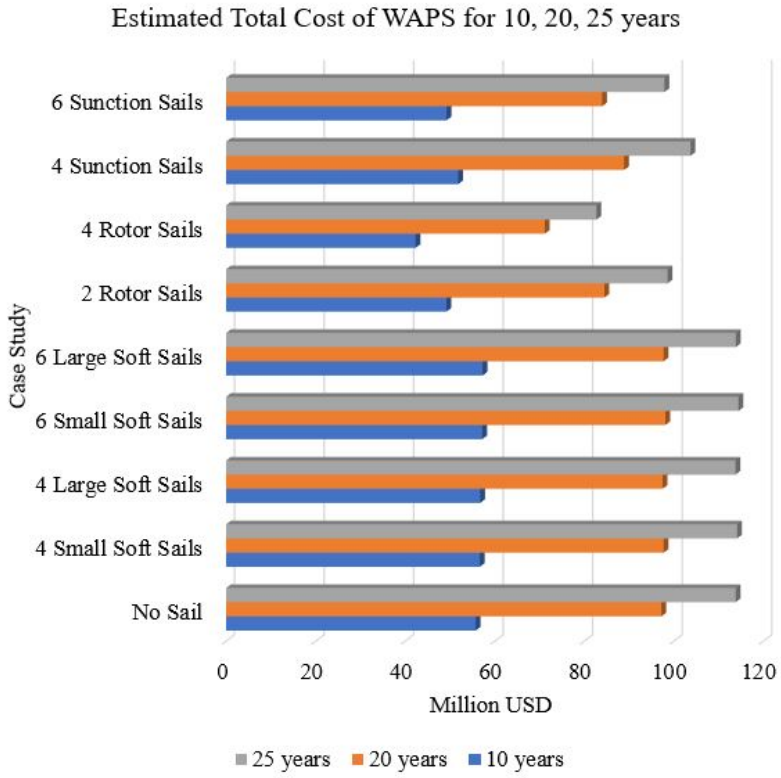


Figure 4: Cost Predictions for each WAPS case study for the first 10, 20 and 25 years respectively.

	Annual Fuel Cost (Million USD)	Annual Compliance Cost (Million USD)	Total Annual Cost (Million USD)
Case 1	4.683	1.823	6.536
Case 2	4.599	1.790	6.469
Case 3	4.556	1.773	6.419
Case 4	4.556	1.773	6.444
Case 5	4.489	1.747	6.367
Case 6	3.907	1.521	5.563
Case 7	3.117	1.213	4.550
Case 8	4.126	1.606	5.842
Case 9	3.836	1.493	5.479

Figure 5: Total Annual Costs for each WAPS case study.